

Over, puppeteer from the US were able to 3D print a prosthetic hand that provided dexterity to people who were either born without fingers or lost them in accidents.

3D printed prosthetics would go on to become a viable alternative to expensive ones made from carbon fiber and titanium alloys. The same technology is currently giving over 50,000 war refugees in Sudan a new lease on life at a cost of about \$100 each.

More recently, 3D printing also gave Dudley, the duckling a new lease on life too. After being attacked by a chicken that left him with just one foot, Doug Nelson, the owner of the shelter where Dudley was brought, in consultation with Terence Loring, founder of a design firm 3 Pillar Designs designed a prosthetic leg. As Dudley would outgrow each limb, Terence would improve the design each time.



The security risk

With the dawning of a new decade in the 21st century, the limits of 3D printing were further stretched. Security, now a growing concern spawned the unmanned aerial vehicle, better known as the drone. They were used in Operation Neptune Spear which led to Osama bin Laden's demise

and China has been using them to spy on us for quite a while. But what half a dozen aeronautical engineers did with a 3D printer proved that the potential that lies hidden inside 3D printing is infinite.

The medical industry was not the only one to embrace 3D printing. Manufacturers of aircrafts and aircraft parts embraced 3D printing to build parts of an unprecedented standard. United Technologies' Pratt & Whitney engine uses vanes and compressors inside its engines made from 3D printed parts. Honeywell 3D prints the heat exchangers for their jet engines and Boeing already make about 300 3D printed parts. Of these parts, some are ducts designed to carry cool air to electronic equipment and because they were had complicated shapes, manufacturing and assembling them were labour intensive. Now those costs have been cut down and more efficient designs are being implemented.